$$\lim_{x\to 0} \exp\left(\frac{1}{x^2 \sin x} \log\left(\frac{x^4 \sqrt[3]{\sin x} - \sin^2 x + x^2}{e^{x^2} - \cos x}\right)\right)$$

$\sim x^3$ $\qquad \sim \frac{3}{2} x^2 \qquad = \frac{x^4/3}{\frac{3}{2}x^2}$

$= 0$

$$x^2 - \sin^2 x = x^2 - \left(x - \frac{x^3}{6} + o(x^3)\right)^2$$

$$= \cancel{x^2} - x^2 + \frac{x^4}{3} + \cancel{\frac{x^6}{36}} + \cancel{o(x^6)} + o(x^4)$$

$$= \frac{x^4}{3} + o(x^4)$$

$$\cancel{x^4 \sqrt[3]{\sin x}} + \frac{x^4}{3} + o(x^4) = \frac{x^4}{3} + o(x^4)$$

$$\frac{x^4 \sqrt[3]{\sin x}}{x^4} = \sqrt[3]{\sin x} \longrightarrow 0 \implies x^4 \sqrt[3]{\sin x} = o(x^4)$$

$$e^{x^2} - \cos x$$

$$e^t = 1 + t + \frac{t^2}{2} + o(t^2) \implies e^{x^2} = 1 + x^2 + \frac{x^4}{2} + o(x^4)$$

$$e^{x^2} - \cos x = 1 + x^2 + \frac{x^4}{2} - \left(1 - \frac{x^2}{2} + \frac{x^4}{4}\right) + o(x^4)$$

$$= \boxed{\frac{3}{2}x^2 + o(x^2)}$$

primo termine significativo

NON SERVE ALTRO